

**Software Design and Analysis
(CS3004)**

Course Instructor(s):

Mr. Majid Hussain , Dr. Javaria Imtiaz, Dr.Afia
Zafar

Section(s): A,B,C,D,E,F,G

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 3

Date: Apr 9, 2026

Roll No

Course Section

Student Signature

Do not write below this line.

Attempt all the questions.

[Question 01]

Part (a): Draw a system sequence diagram for the below main success scenario. [5 marks]

Functional Requirement:

Place Order with Multiple Items and Payment Processing

Main Success Scenario

1. The customer opens the app and selects a restaurant.
2. The system displays the menu items of the selected restaurant.
3. The customer selects multiple food items.
4. (This step repeat three to 4 times for adding multiple items)
5. Customer proceeds to checkout.
6. The system calculates the total bill including delivery charges and taxes.
7. The customer selects the payment method (Cash / Card / Wallet).
8. System processes payment:
 - a. If payment is successful, proceed to the next step.
 - b. Else, the system asks the customer to retry payment or choose another method.
9. The system confirms the order and assigns a delivery rider.
10. The system sends order confirmation notification to the customer.

Part (b): Draw a class diagram to represent the following description of domain objects involved in developing a registration system for voters in provincial elections: "Each person can be either an eligible voter or an ineligible voter. Eligible voters are registered in only one electoral district (known as a 'riding'), and for each registration instance, details such as the date of registration and current status need to be stored. Eligibility is determined by address. Each person can be assumed to have one current address, and each address is in either a town or city. Some ridings contain several towns, while some large cities may contain more than one riding." Be sure to include names and multiplicities on all associations, and state any assumptions you make. How would you modify your diagram to handle the following situations: (a) we may need to retain information on people who no longer live in this

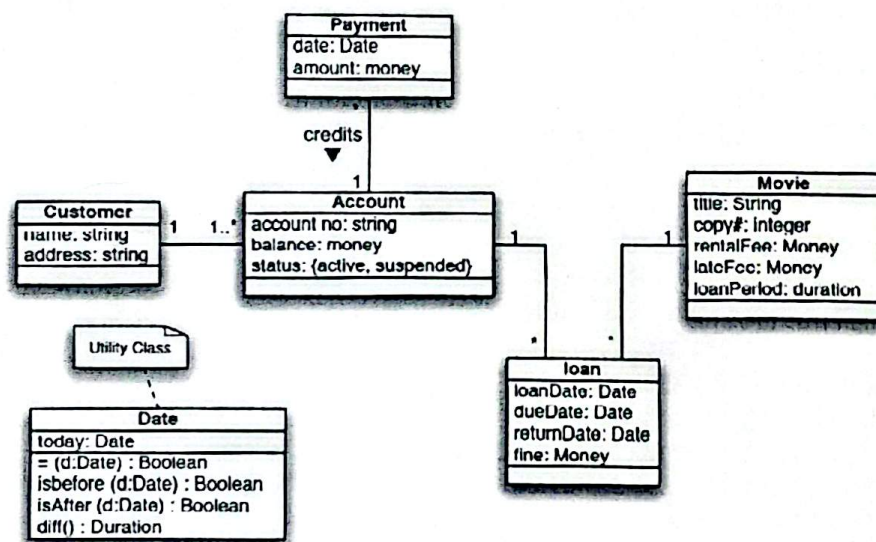
National University of Computer and Emerging Sciences

Islamabad Campus

province; (b) some people own property in more than one riding, and hence apply for registration in more than one riding; (c) people with more than one property are required to designate one as their 'primary residence' to determine which riding they can vote in. [7+3=10 marks]

[Question 02]

Part (a): The following class diagram shows part of the design of software for EasyDVD, a web-based DVD rental agency. EasyDVD allows customers to browse its website to see the inventory of DVDs, and select DVDs for rent. DVDs are sent to the customer and returned by mail. Each DVD has a set rental fee and a set loan period. [10 marks]



1. There are no associations shown for the Date class. How would you redraw the diagram to show such associations? What are the advantages and disadvantages of adding these explicitly? [2.5 marks]
2. EasyDVD wants to restrict customers to have no more than 3 DVDs on loan at a time. How would you modify the diagram to show this constraint? Note: EasyDVD still needs to keep track of (a potentially large number of) past loans for each customer's account. [2.5 marks]
3. EasyDVD wants to simplify its pricing structure, so that there are a small number of categories of movies, where each category has a set rental fee and loan period. How would you modify the diagram to allow this? [2.5 marks]

4. EasyDVD wants to add a new subscription service in addition to the existing ad hoc rental service. In a subscription service, customers maintain an ordered lists of movies they would like to rent, and whenever they return a movie, the next movie on the list is sent out. For the subscription service, customers pay a fixed monthly fee rather than individual rental fees on each movie, and each rental is open-ended – i.e. there is no due date and no fine. How would you modify the class diagram to facilitate this service? [2.5 marks]

Part (b) Sketch a UML Sequence Diagram for the process of renting a DVD for the EasyDVD service described at the start of the previous question (i.e. the regular rental, not the subscription service). Be sure to select a DVD to rent, and show how the system sets up a new loan with the appropriate dates set, using the Date class to calculate these. Please consider the GRASP pattern while designing this SD. Mention the type of pattern you have used. [5 marks]

[Question 03]

Consider a **Vending Machine System** where a Customer interacts with a Vending Machine to purchase an item. The interaction is defined by the following behavioral logic:

1. The Customer (Actor) initiates the process.
2. The Vending Machine immediately updates its internal state by calling its own private method.
3. The system must then evaluate a conditional **Alternative (alt)** fragment:

Provide the Java implementation for all classes involved, following the standard mapping of UML fragments (Self-calls and Alt frames) to code. [10 marks]

